Colour-blind Series: Anomalous Trichromacy

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What is it for?

All three types of colour sensing cells are present, but one of them doesn't function as strongly as it should. The result is weakened rather than absent colour vision.

- **Protanomaly** is a reduced sensitivity to red, making it harder to distinguish red from green or brown
- **Deuteranomaly** is a reduced sensitivity to green. It is the most common form of colour vision deficiency overall
- **Tritanomaly** is a reduced sensitivity to blue, making blue and green, and yellow and red, harder to tell apart

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

Useful Links

WCAG 1.4.1: Use of Colour
Colour Blindness Simulator (Coblis)
Colour and Accessibility (WebAIM)